

AICHU TAN

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PROFESSIONAL SUMMARY

Data Scientist with experience building and deploying end-to-end machine learning and AI systems across public health, agriculture, and urban analytics. Specialized in time-series forecasting, computer vision (YOLO), and LLM-powered applications (LangChain, RAG). Proven ability to develop production-ready solutions using Streamlit, Hugging Face, and AWS, with strong expertise in feature engineering, model evaluation, and translating complex data into actionable insights.

TECHNICAL SKILLS

Programming: Python, SQL, R, JavaScript, C++

Machine Learning: XGBoost, Random Forest, Scikit-learn, Clustering, K-Prototypes, Feature Engineering

Deep Learning & Time Series: LSTM, ARIMA, SARIMAX, Exponential Smoothing, TensorFlow, YOLOv8

Data & Visualization: Advanced SQL (Joins, CTEs, Window Functions), Pandas, NumPy, Tableau, Streamlit, Matplotlib, ArcGIS

Cloud & MLOps: AWS (EC2, S3, RDS), MLflow, Git, Hugging Face Spaces, Supabase, Render

PROFESSIONAL EXPERIENCE

Data Science Researcher - SDSU DiMoLab

Sep 2025 - Present

- Designed and implemented end-to-end SQL and Python ETL pipelines to ingest and process 27+ years of high-volume longitudinal data, standardizing formats across meteorological, entomological, and population datasets.
- Engineered automated data validation workflows to flag out-of-spec anomalies, establishing baseline data-quality rules to filter out reporting noise before downstream modeling.
- Wrote advanced SQL queries utilizing complex joins, window functions, and CTEs to extract and prepare data models for time-series forecasting.
- Built time-series statistical models to predict outbreaks, optimizing data granularity into 3-day and 5-day windows to achieve an R^2 up to 0.89.
- Co-authoring a research manuscript for submission to a peer-reviewed journal.

AI Researcher - SDSU Metabolism of Cities Living Lab

Feb 2025 - Dec 2025

- Developed an LLM-powered retrieval system (RAG, LangChain) utilizing specialized research PDF documentation to generate grounded, evidence-based agricultural recommendations with minimized hallucinations.
- Executed data-quality root cause analysis using confusion matrices to diagnose misclassifications across 35,000 plant disease images.
- Overhauled dataset annotation strategy by establishing strict quality control and re-annotation rules for 13,500+ images, scaling model performance (YOLOv8) from 70% to 89.4% mAP@50.
- Deployed a production-ready web interface using Streamlit and Hugging Face Spaces to bridge user interaction with underlying ML models. Co-authored peer-reviewed research accepted at the 4th International One Health Conference (Rome, 2025).

SELECTED PROJECTS

Predicting Restaurant Ratings Using Yelp Metadata | MLflow, XGBoost, RF, Pandas

- Built an end-to-end machine learning classification pipeline using 36,000+ Yelp records to evaluate features predicting high vs. low consumer ratings.
- Engineered 61 distinct business metadata features spanning service types, operational hours, and accessibility metrics, tracking all experiments via MLflow.
- Utilized K-Prototypes clustering to uncover consumer archetypes, discovering that service accessibility metrics drive positive ratings more robustly than premium pricing tiers.

Online Ticket Reservation System | MySQL, SQL, Python, AWS RDS

- Designed and implemented a normalized relational database system through advanced ERD and data modeling.
- Built automated business rules and validation logic, including trigger-based checks, to ensure transactional data integrity and prevent invalid entries.
- Developed a Python dashboard using Datapane to visualize transactional trends, revenue drivers, and customer behavior patterns.

Mapping for Change - Urban Inequality Analysis | R, Tableau, Spatial Analysis, Time Series, Data Visualization

- Built an interactive ArcGIS dashboard to analyze homelessness, housing affordability, and environmental inequality across Downtown San Diego.
- Integrated 5+ public datasets (ACS, 311, PITC, Tree Equity Score) and applied spatial analysis (heatmaps, choropleths) to identify structural inequality clusters.
- Uncovered key statistical relationships between poverty, rent burden, and reduced tree equity, highlighting areas of environmental injustice to support data-driven urban planning.

OTHER EXPERIENCE

Business Owner - Wholesale Clothing

Apr 2009 - Oct 2016

- Increased sales by 50% through data-driven market analysis and customer segmentation
- Optimized supply chain operations using forecasting and analytics
- Managed international import/export operations across Taiwan, China, and Indonesia

Process Engineer - ProMOS Technology

Jul 2004 - Feb 2009

- Improved DRAM yield via **statistical process control (SPC)** and root cause analysis.
- Reduced manufacturing defects through data modeling and process optimization.

EDUCATION

M.S. in Big Data Analytics, San Diego State University - GPA 4.0

Aug 2024 - Dec 2025

Google Data Analytics Professional Certificate

Dec 2022

B.S. in Chemical Engineering, National Cheng Kung University (Taiwan)

Sept 1999 - June 2003